

Kabilan S

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Embedded Systems and Hardware Engineer with experience in circuit design, PCB development, firmware programming, and system-level debugging. Skilled in working with ESP32, Raspberry Pi, Arduino, and various communication protocols, including UART, I2C, SPI, and RS232. Demonstrated ability to translate ideas into functional prototypes through strong analytical thinking, hands-on experimentation, and collaborative development. Highly motivated, adaptable, and committed to continuous learning in research-driven environments.

Experience

MARCH 2025 – PRESENT

Hardware Engineer | Jaswa Vending Private Limited | Chennai, Tamil Nadu

- Led the **end-to-end development** of an automated perfume vending machine, from concept to fully assembled and functional product.
- Designed the complete **3D mechanical structure and enclosure** using Fusion 360, optimized for manufacturability and maintenance.
- Worked with communication protocols such as UART, I2C, SPI, and MQTT for device-to-device and cloud communication.

FEBRUARY 2024 – APRIL 2024

Research Intern | IIT Roorkee | Roorkee, Uttarakhand

- Worked as a Research Intern on a **government-funded R&D project**, contributing to embedded systems and electronics design.
- Performed **reverse engineering of electronic circuits**, including analysis of signal conditioning and interface stages.
- Contributed to **application development** for data visualization, logging, and system interaction.

Skills

• Embedded C/C++ • ESP32, Raspberry Pi, Arduino • PCB & Circuit Design • IoT Prototyping • Fusion 360 (3D Design) • Hardware – Software Integration • Sensor & Actuator Integration • Team Player

Education

MAY 2025

Bachelor of Engineering - (Electronics & Communication Engineering) | Rajalakshmi Institute of Technology | Chennai, Tamil Nadu

8.65 CGPA • First class with distinction.

Activities

• Reading • Entrepreneurship Development • Cricket • Podcasting • Singing • Travel